

AF3Score: A Score-Only Adaptation of AlphaFold3 for Biomolecular Structure Evaluation

Yu Liu, Qilin Yu, Di Wang, and Mingchen Chen*



Cite This: *J. Chem. Inf. Model.* 2025, 65, 8207–8214



Read Online

ACCESS |



Metrics & More

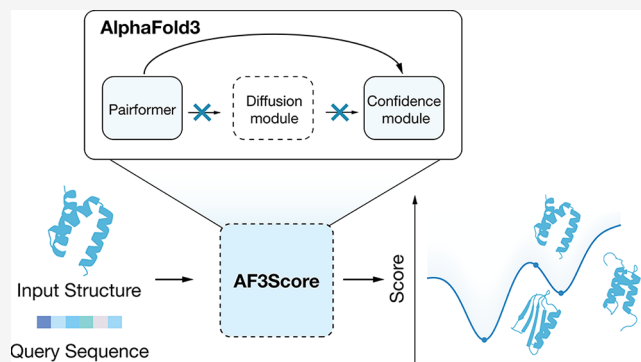


Article Recommendations



Supporting Information

ABSTRACT: Scoring biomolecular complexes remains central to structural modeling efforts. Recent studies suggest that AlphaFold (AF) – a revolutionary deep learning model for biomolecular structure prediction – has implicitly learned an approximate biophysical energy function. While many researchers highly rely on AF-derived scores for structure evaluation, existing AlphaFold2-based implementations require iterative refinement of the input structure, leading to biased scoring. To address this limitation, we adapted AlphaFold3 into a score-only model, AF3Score, by directly feeding input coordinates into the confidence head while bypassing the diffusion-based structure module. AF3Score demonstrates robust performance in structural quality assessment across diverse systems, including monomeric proteins, protein–protein complexes, de novo designed binders, fold-switching proteins, and protein–ligand complexes. In benchmarking designed binder screening, AF3Score outperformed state-of-the-art methods for 8 out of 10 targets. Moreover, combining AF3Score with AlphaFold2-derived methods significantly improved the enrichment of experimentally validated binders, increasing the success rate from 15.2 to 31.6%. Additionally, AF3Score effectively identified stable conformations in fold-switching proteins, whereas AlphaFold predominantly predicted only the dominant fold. These findings highlight the broad applicability of AF3Score, from high-throughput screening in de novo binder design to filtering docking-generated poses and molecular dynamics (MD) trajectories.



INTRODUCTION

Recent advances in biomolecular structure prediction have revolutionized drug discovery, enabled rational protein design^{1,2} and deepened our understanding of molecular interactions. At the forefront of this progress are deep learning-based approaches such as AlphaFold2³ and RosettaFold,⁴ which achieved remarkable accuracy by leveraging coevolutionary information from multiple sequence alignments (MSAs). These evolutionary constraints effectively reduce the conformational search space by encoding structural conservation patterns across homologous sequences. However, the predictive performance of these models deteriorates significantly in the absence of MSAs.^{3,5}

The folding landscape of a protein is uniquely determined by its amino acid sequence, with evolutionary pressures sculpting this landscape to ensure proper folding.^{6,7} While an optimally folded protein requires a funnel-shaped and minimally frustrated energy landscape, functional constraints often introduce local frustrations that hinder efficient conformational search.⁸ This raises an important question: How does AI navigate the rugged folding landscape? One plausible hypothesis is that while AlphaFold2 has learned an accurate energy function for structure evaluation, MSA-derived coevolutionary information is essential for efficiently searching for an approximate global minimum.⁹ This aligns with AlphaFold2's

prediction scheme: an initial MSA-guided search to identify plausible structural templates, followed by iterative refinement using learned energetic potentials.

This concept is further supported by AF2Rank,⁹ which demonstrated that AlphaFold2's confidence metrics correlate with structural accuracy even when MSAs are omitted, suggesting the potential of AF2 as a generalizable energy function. Additionally, the metrics show reasonable correlation with experimental success in both monomer and complex structure predictions.¹⁰

However, existing AlphaFold2-based scoring methods cannot directly evaluate a given structure; instead, they require iterative perturbations of the input coordinates. In contrast, AlphaFold3 exhibits a reduced reliance on MSAs,^{3,5} while achieving improved prediction accuracy. To harness this learned potential, we developed AF3Score, a score-only evaluator that adapts AlphaFold3 by directly inputting

Received: March 28, 2025

Revised: July 7, 2025

Accepted: July 7, 2025

Published: July 17, 2025



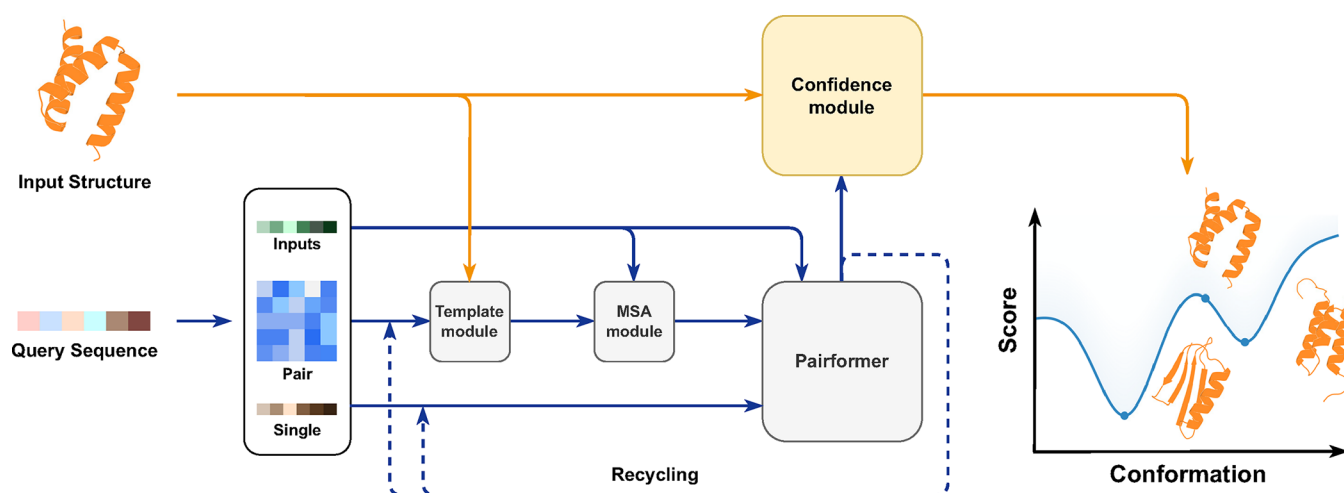


Figure 1. Overview of the AF3Score framework. Schematic representation of the AF3Score pipeline shows processing modules (rectangles) and data flow (arrows) in the streamlined architecture. Yellow arrows indicate the integration of the input structure, and blue arrows represent network activations. The conformational energy landscape was illustrated with fold-switching protein pairs GA98/GB98 (PDB: 2LHC/2LHD) and a computationally generated decoy from GA98.

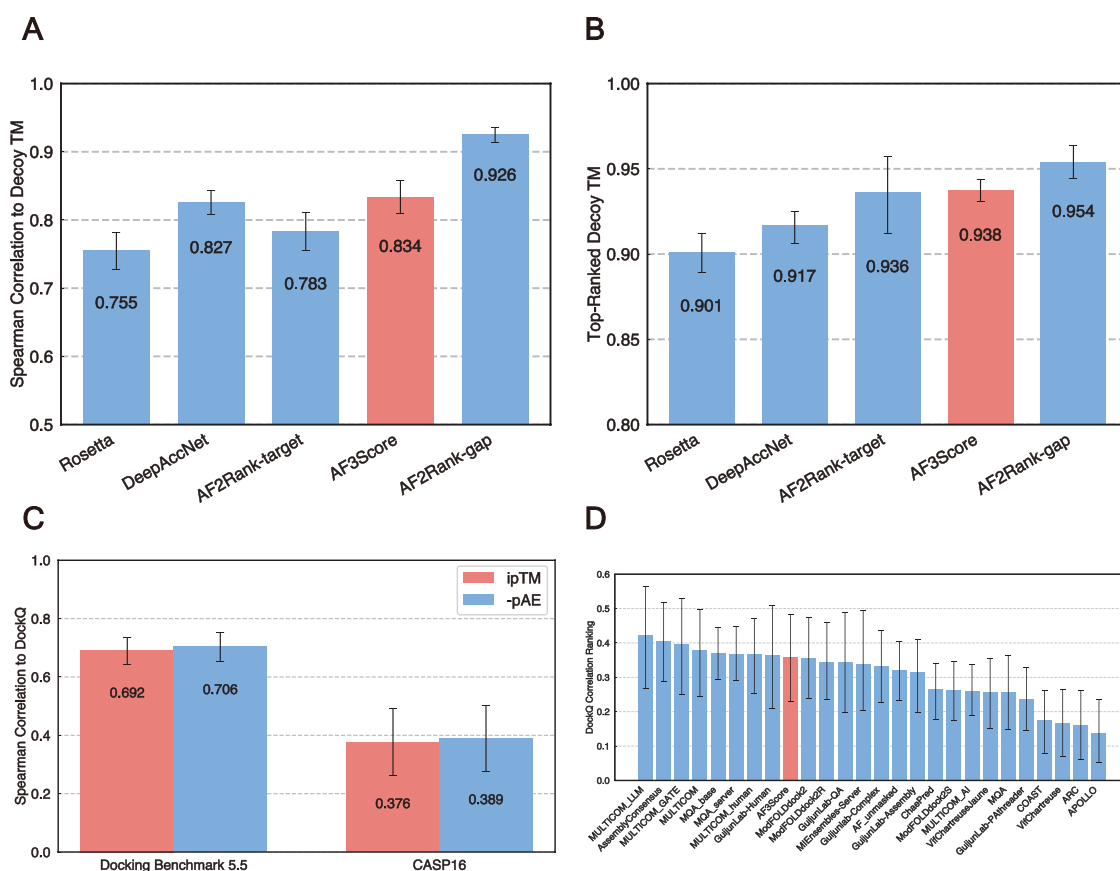


Figure 2. (A, B) Quality assessment of protein structures on the ROSETTA decoy data set. (A) Spearman correlations between different scoring methods and TM-scores of the decoy docking ensemble. (B) Mean TM-scores of the top-ranked decoys by each method. AF3Score is highlighted as red. (C, D) Evaluation of AF3Score on two PPIs data sets. (C) Spearman correlations between metrics from AF3Score and DockQ values on two data sets are shown. (D) Rankings of assessment of interface quality by 25 methods from CASP16. All error bars represent bootstrap 95% confidence intervals of the mean.

structure coordinates as templates and bypassing the diffusion module. We systematically validated AF3Score across four critical benchmarks: (1) ranking decoys of protein monomers, (2) assessing protein–protein complexes (both antibody and nonantibody), (3) evaluating fold-switching proteins, (4)

assessing binder enrichment in de novo design, and (5) assessing protein–ligand complexes. Our results demonstrate AF3Score’s superior correlation with experimental outcomes, highlighting its potential as a valuable tool for practical applications and drug discovery.

RESULTS

Adapting AlphaFold3 to a Score-Only Model. AF3Score transforms AlphaFold3 into a structure evaluator through two key modifications (Figure 1). First, the genetic search and template search blocks are removed, eliminating reliance on MSAs and external structural databases while preserving the conformer generation component, which is essential for handling ligands and amino acid side chains. Second, the diffusion module is bypassed, allowing the input coordinates to serve dual roles: they act as structural templates within the template module feeding into the pair representation, and as the evaluation target directed to the confidence head. In this process, the atomic coordinates of the input structure are integrated with processed 1D and 2D information from the pairformer to generate quality metrics.

This approach differs from AF2Rank, which derives confidence metrics by combining AlphaFold2's inherent scoring with structural alignment between input and predicted models. By directly scoring the input structure itself, AF3Score provides a more immediate and unbiased assessment of the structural quality.

Benchmarking AF3Score on Monomers and Protein–Protein Interactions. AlphaFold3 (AF3) demonstrates slightly improved performance in monomeric protein structure prediction compared to AlphaFold2 (AF2). To evaluate whether AF3Score captures a more accurate approximation of the folding landscape, we assessed its ability to rank decoy structures relative to their native conformations. This benchmarking was performed using the Rosetta decoy data set,¹¹ which comprises 133 single-chain targets and approximately 180,000 decoys.

AF3Score was compared against several established methods, including Rosetta,¹² DeepAccNet,¹³ and two versions of AF2Rank (target and gap modes).⁹ AF3Score achieved a Spearman correlation of 0.834 with TM-scores across all test cases, surpassing Rosetta (0.760) and DeepAccNet (0.831) (Figures 2A, S1, and S3, and Table S1). Notably, AF3Score successfully identified near-native structures with its top-ranked predictions averaging a TM-score of 0.938 (Figure 2B). These results establish AF3Score as a useful tool for structural quality assessment, although AF2Rank remains slightly superior for this specific task.

Accurately characterizing protein–protein interactions (PPIs) is essential for understanding cellular functions and drug discovery. Classical computational approaches such as molecular docking^{14,15} play a crucial role in complementing experimental methods, particularly in evaluating interface quality and identifying near-native conformations from computationally generated ensembles.^{16,17} To assess AF3Score's capability in PPI evaluation, we tested it on two complementary data sets: the Docking Benchmark 5.5,¹⁸ which covers a diverse range of seen interactions for AlphaFold3, and the CASP16 quality assessment (QA) category, which focuses on generalization to novel interactions.¹⁹ On the Docking Benchmark 5.5 data set, AF3Score demonstrated strong discrimination of near-native structures against decoy structures from docking simulations (Figures 2C, S2, S4, and S5, and Table S2). In CASP16 QA Mode 1 (QMODE1), AF3Score ranked ninth among all participating groups (Figures 2D, S2, S6, and S7, and Tables S3 and S4). Considering that all of the top-ranking methods are AF3-based and computation-

ally expensive, AF3Score achieves comparable performance with improved computational efficiency (Table S6).

AF3Score Accurately Identifies Diverse Binding Modes. We analyzed several representative cases to evaluate the ability of AF3Score to distinguish between different binding modes and recognize distinct epitopes. In the complexes formed between TEM-1 β -lactamase and β -lactamase inhibitor protein (BLIP) or β -lactamase inhibitor protein-II (BLIP-II), although BLIP and BLIP-II adopt completely different folds, they both exhibit a competitive mode of inhibition toward TEM-1.²⁰ In this case, AF3Score accurately recognized both binding modes (Figure 3A,B).

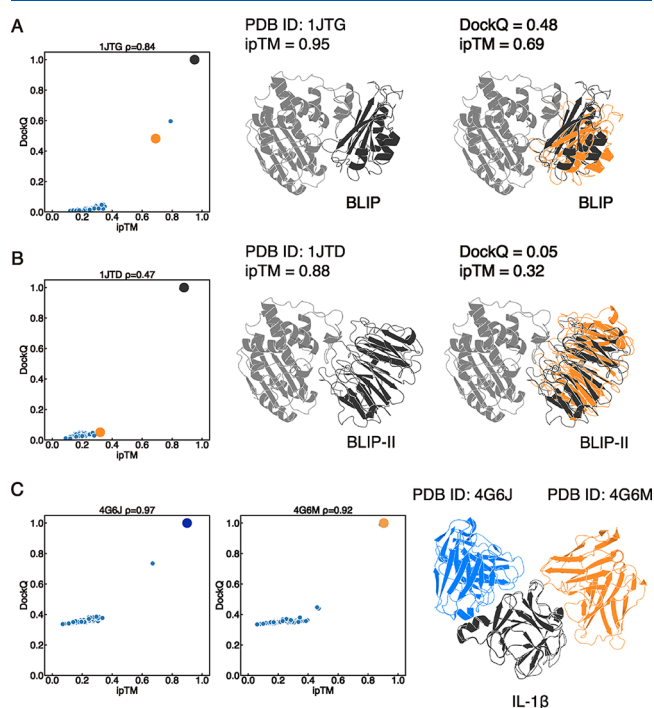


Figure 3. Case studies from the evaluations of AF3Score on PPIs. (A, B) Structural superposition of BLIP and BLIP-II complexes with TEM-1 β -lactamase (gray), showing native inhibitor structures (black) and decoy conformations (orange). Corresponding correlation plots display AF3Score values for native (black) and decoy (orange) structures. (C) Structural comparison of IL-1 β binding to two different antibodies with two distinct epitopes: canakinumab (PDB ID: 3G6J, blue) and gevokizumab (PDB ID: 3G6M, orange).

Another example illustrates the ability of AF3Score to identify distinct epitopes of the same antigen. Interleukin-1 β (IL-1 β) is a key orchestrator in inflammatory and immune responses.²¹ Two highly specific IL-1 β monoclonal antibodies, Canakinumab and gevokizumab, exhibit divergent binding mechanisms. Canakinumab competitively inhibits IL-1 β association with its receptor due to steric overlap between their binding interfaces on IL-1 β . In contrast, gevokizumab occupies an allosteric site on IL-1 β , forming a complex that reduces the binding affinity of IL-1RI. In this case, AF3Score accurately identified both epitopes, demonstrating the capability of AF3Score to recognize different local structural information and thereby distinguish between distinct binding modes (Figure 3C).

Despite its reasonable performance, AF3Score faces challenges in capturing certain novel interfaces, and this might be due to a lack of more interaction data during training. For example, in the caspase-9:BIR3 complex, the confidence

scores of docking decoys fall within a narrow range (less than 0.2), and the native structure's score is only marginally higher than those of the decoys (Figure 4).

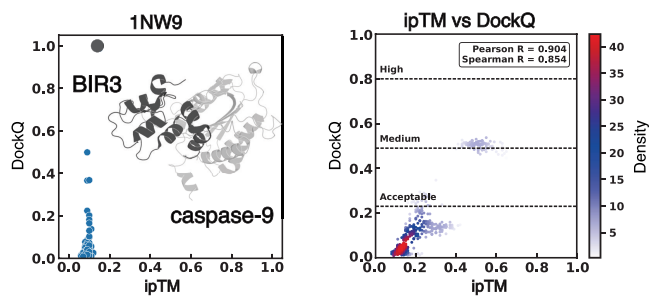


Figure 4. Analysis of a challenging case: the BIR3-caspase-9 complex. Left: Correlation between AF3Score ipTM and DockQ scores, with HDock-generated decoys shown in blue and the native structure in black. The structural representation shows the native complex with BIR3 in black and caspase-9 in gray. Right: Distribution of AF3-sampled structures (500 structures from 100 seeds, 5 structures per seed) plotted as ipTM versus DockQ scores, with sampling density indicated by color intensity. Horizontal lines mark quality thresholds for complex structures (Acceptable: 0.23, Medium: 0.49, High: 0.8).

AF3Score Enhances the Success Rates of De Novo Protein Binder Design. The design of protein binders faces two primary failure modes: (1) the designed sequence fails to adopt its intended monomeric conformation, and (2) the designed sequence correctly folds but does not bind the target.¹⁰ Previous work showed that AF2-initial-guess offered a promising solution, outperforming physically based methods like Rosetta in predicting experimental successes.¹⁰

Using a curated subset from a published data set of in silico-designed binders,¹⁰ AF3Score demonstrated strong discriminative power across ten distinct targets (Figures S5B,C and S8). Compared to existing methods—including AF2-initial-guess,¹⁰ RoseTTAFold2,^{22,23} DeepAccuracyNet,¹³ and Rosetta,¹² AF3Score achieved superior performance in 6 out of 10 cases. Notably, AF3Score matched or outperformed AF2-initial-guess, the current state-of-the-art for this task, in 8 of 10 targets (Figure 5C). A significant improvement was observed when combining AF2-initial-guess with AF3Score, identifying 98 candidates through intersectional selection, with 31 (31.6%) experimentally validated hits. This approach led to a 2-fold increase in success rate, improving from 15.2% (AF2-initial-guess) to 31.6% (Figure 5D). AF3Score was further evaluated in a recent competition hosted by Adaptic Bio,²⁴ which aimed to design binders targeting the epidermal growth factor receptor (EGFR). Compared to AF2, AF3Score demonstrated superior enrichment capabilities for successful designs. In modern protein design, monomeric metrics serve as an initial filter for structural quality and ensure that the designed sequences are foldable before evaluating binding functionality. Using this approach, AF3Score identified 75% (3/4) of successful binders within its top 1% ranked designs, whereas AF2 captured only 25% (1/4) (Figure 5E). Using complex metrics, the ipTM from AF3Score achieved a validation accuracy of 50% (2/4), whereas AF2's PAE interaction metric failed to identify any true binders (0/4) (Figure 5F). Notably, a consensus filtering approach that combined the top 5% of candidates from both AF3Score and AF2 achieved a success rate of 55.6% (5/9 validated). However, direct correlation analysis between AF3Score and binding affinity revealed

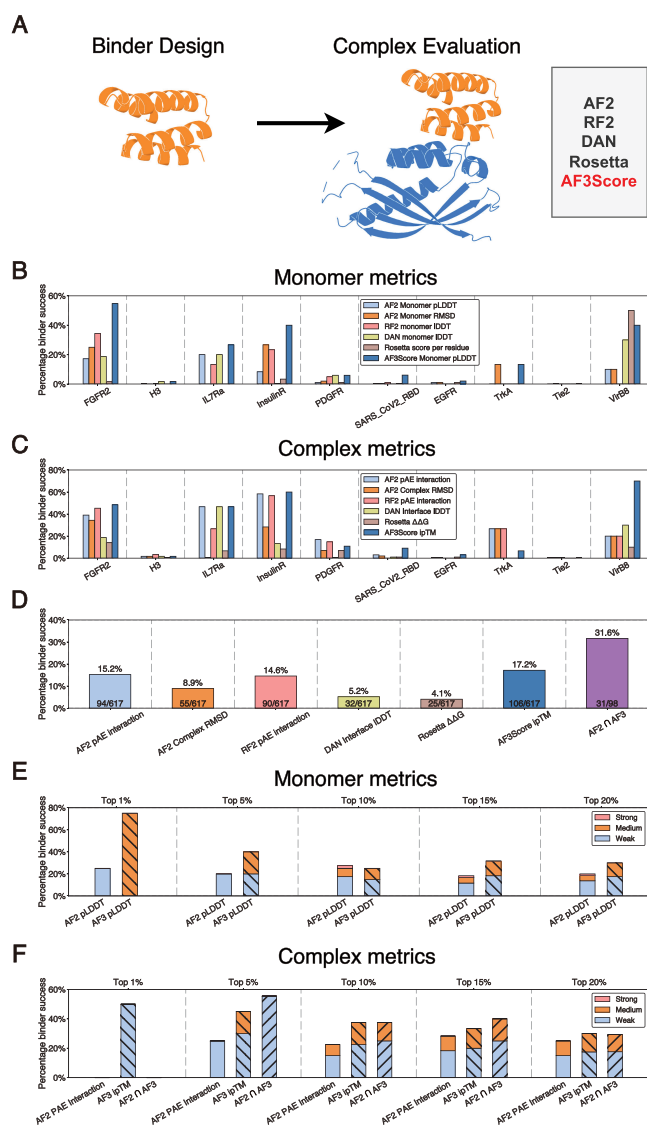


Figure 5. Benchmarking on the EGFR binder design contest by Adaptic Bio. (A) Schematic illustration of different methods used to assess designed protein complexes, with mini-binders shown in yellow and target proteins in blue. (B–D) Evaluation of designed protein binders across 10 targets from Cao et al. (B, C) Comparison of experimental success rates (YSD SCS0 < 4 μ M) for the top 1% of designs selected based on different monomeric (B) or protein complex (C) metrics. (D) Overall success rates of the top 1% of designs selected using individual protein complex metrics, as well as the intersection of AF2 pAE interaction and AF3Score ipTM (AF2 $\cap$ AF3). (E, F) Evaluation of designed protein binders targeting EGFR. Experimental validation identified a success rate of 13.25% (53/400 designs). (E, F) Comparison of experimental success rates for designs selected based on different monomeric (E) or protein complex (F) metrics. Bars indicate the percentage of successful binders within different binding strength categories (strong, medium, and weak) across the top 1, 5, 10, 15, and 20% of structures ranked by each metric. Different hatch patterns within bars distinguish between the evaluation methods.

consistently weak relationships across both de novo designed binders and natural protein–protein complexes, confirming that AF3Score functions primarily as a structural quality filter rather than a quantitative binding affinity predictor (Figure S16).

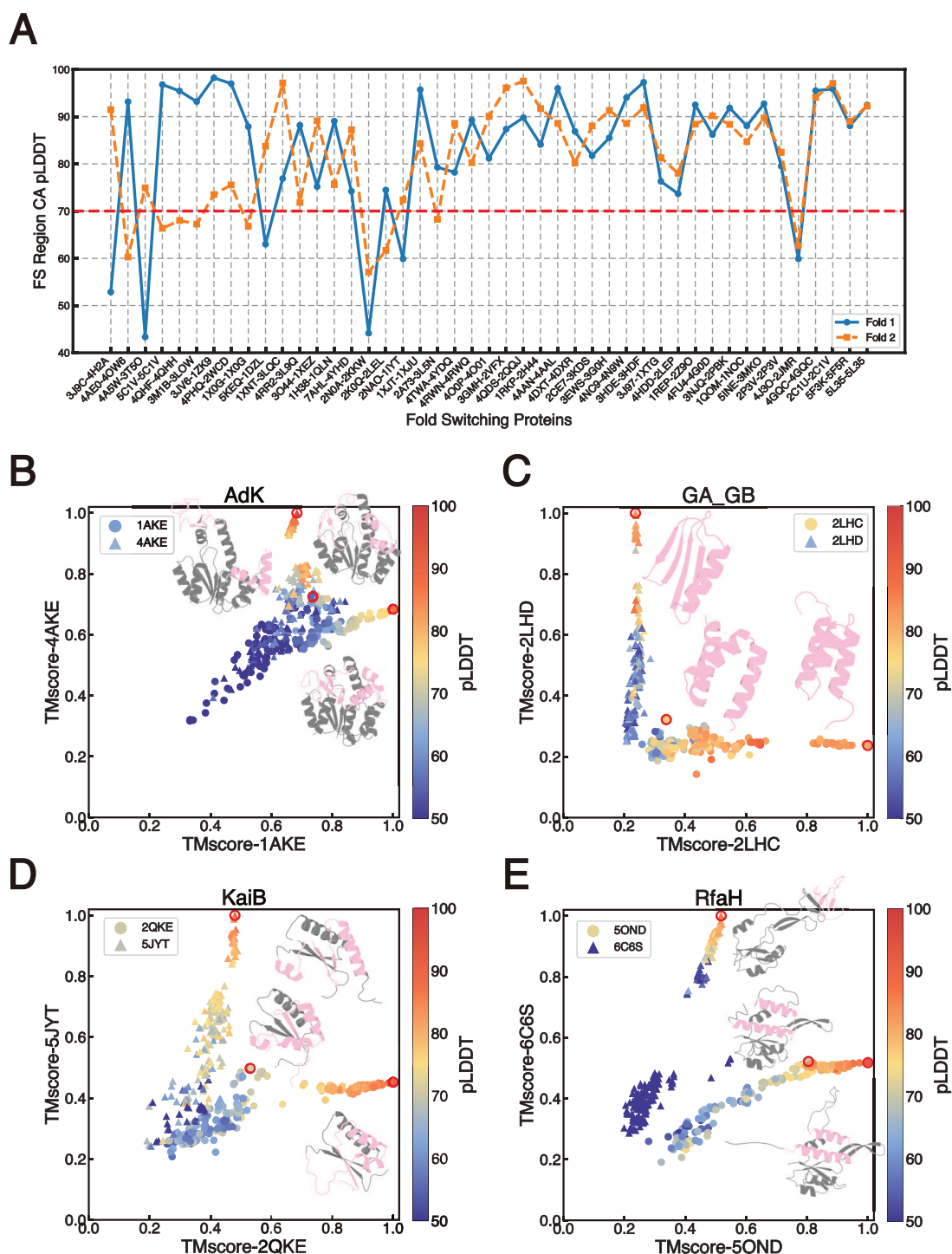


Figure 6. Benchmarking AF3Score on fold-switching proteins. (A) Confidence scores assigned by AF3Score for 45 pairs of fold-switching proteins. The two native conformations are colored orange and blue. The red line indicates a fold-switching (FS) region pLDDT threshold of 70, marking high-confidence predictions in these regions. (B–E) Analysis of the representative fold-switching proteins. The x- and y-axes represent TM-scores relative to each native fold. Each point corresponds to a structure, colored according to the C α pLDDT score of its fold-switching region. Triangles and circles distinguish the two alternative folds. Native structures and selected decoys (highlighted by red circles) are shown, with fold-switching regions in pink and the remaining structure in gray.

Benchmarking AF3Score on Fold-Switching Proteins.

Fold-switching proteins adopt two or more distinct folded states from a single amino acid sequence, undergoing either spontaneous conformational interconversion or stimulus-responsive structural transitions.^{25,26} Predicting alternative folds remains a major challenge for AlphaFold2 (AF2).²⁷ A recent benchmark²⁸ evaluating all AlphaFold versions,

including AF2,³ AF2-multimer,²⁹ AF3,⁵ and enhanced sampling methods,^{30,31} demonstrated AF2's limitations in predicting fold-switching events. While multiple sequence alignments (MSAs) provide valuable evolutionary context for dominant protein structures, they often fail to capture alternative conformations.^{30,32} For a comprehensive scoring function, both dominant and alternative conformations should

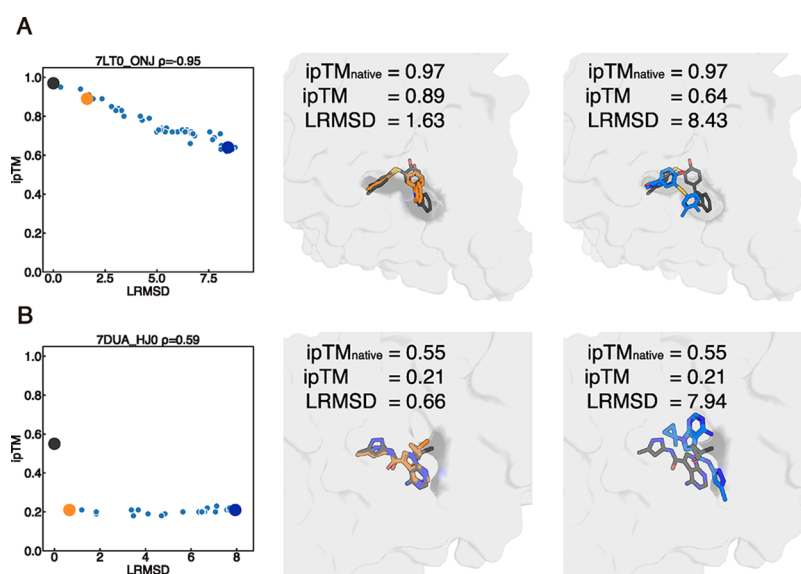


Figure 7. Case studies of the evaluation of AF3Score on the protein–ligand complex data set. (A, B) Views of ligand interactions with protein pockets (gray surface) for 7LT0 and 7DUA complexes. Correlation plots show AF3Score’s ipTM versus LRMSD values, with representative structures highlighting near-native (yellow) and high-LRMSD (blue) poses. While AF3Score correctly identified the native pose (ipTM = 0.55), it failed to properly rank near-native decoys. Specifically, poses with LRMSD of 0.66 Å received the same low scores (ipTM = 0.21) as clearly incorrect poses (LRMSD = 7.94 Å).

achieve high confidence scores. To test this hypothesis, we reframed the fold-switching prediction as a structural scoring problem.

Using an established database of fold-switching proteins,²⁷ we curated 45 protein pairs, excluding cases with insertions, deletions, or mutations in their fold-switching regions. Initial evaluation using AF3Score (Figure 6A) revealed comparable confidence scores for both conformations, with mean values of 83.15 and 81.85 for Fold 1 and Fold 2, respectively, unlike the existing methods, which often exhibit strong bias toward a single conformation.

To further assess AF3Score’s ability to capture fold-switching, we examined the conformational landscapes of several well-characterized fold-switching proteins (Figures 6B–E and S9–S11). While auxiliary approaches, such as local energetic frustration analysis, can help AF2 sample alternative states,³³ we investigated whether AF3Score inherently overcomes this limitation. For each of the six cases, 200 structures were sampled using 3D-Robot between the two alternate states, and the resulting structure ensemble was later scored.

A particularly illustrative example is Adenylate kinase (AdK) (Figure 6B), a multidomain protein that serves as a model for allosteric regulation.³³ While certain decoy structures exhibited moderate structural similarity to both native folds (TM-score > 0.7), their AF3Score confidence values were notably low, highlighting their sensitivity to fold-specific features. To further evaluate the robustness of AF3Score in alternative state prediction, we analyzed AdK conformational ensembles generated via MD simulations.³³ The results (Figure S15A,B) demonstrate that AF3Score successfully identifies stable conformations along the reaction pathway. These results demonstrate the ability of AF3Score to accurately identify stable, experimentally consistent conformations in fold-switching regions.

The GA/GB system (Figure 6C) presents another fascinating case where a single amino acid substitution (L45Y) leads to dramatically different folds: GA adopts a

three-helix bundle while GB forms a $4\beta + \alpha$ structure. The distinct pattern observed in the GA/GB analysis, characterized by low TM-scores between native structures, underscores AF3Score’s ability to distinguish such drastically different yet related folds.

KaiB is a protein that regulates cyanobacterial circadian rhythms through its fold-switching mechanism.³⁴ Within the fold-switching landscape of KaiB’s superfamily, members are generally expected to adopt a predominant thioredoxin-like fold. However, those capable of switching folds also exhibit an alternative ground state. Unlike other methods that predict only the dominant thioredoxin-like fold,^{27,35} AF3Score successfully identifies both conformational states (Figure 6D). Similar observations were found in RfaH (Figure 6E), a transcription factor that regulates the expression of virulence genes in *E. coli*, which possesses a C-terminal domain (CTD) capable of undergoing a complete transition between α -helix and β -sheet folds.

Moderate Performance of the AF3Score on Protein–Ligand Complexes. We further extended AF3Score to evaluate protein–ligand complexes, where it exhibited moderate discriminative power (Figures 7A and S12). Our analysis revealed that AF3Score successfully identified native protein–ligand interactions in 88.2% of cases (157/178), where native structures received higher scores than their corresponding decoys (Figure S13A). However, its performance varied substantially across different proteins - for example, ALK tyrosine kinase receptor showed a strong negative correlation (−0.779) between AF3Score and LRMSD, while other proteins exhibited weaker correlations (Figure S13B). Notably, when examining the relationship with binding affinity using the PDBbind database ($n = 498$), AF3Score showed limited correlation with experimental binding strength (Spearman correlation = 0.140, Figure S14A), and only marginal ability to discriminate between strong ($K_d \leq 10$ nM) and weak binders ($K_d > 10$ μ M) (Figure S14B). A limitation is its reduced accuracy in ranking near-native ligand poses, likely

due to the lack of template-guided refinement for ligand coordinates during representation updates in AlphaFold3 (Figure 7B).

CONCLUSIONS

Building upon the biophysical insights captured in AlphaFold3's energy function, we introduce AF3Score, a scoring framework for evaluating biomolecular structure. This score-only model demonstrates broad applicability across multiple tasks, including the assessment of monomeric protein, protein–protein interaction, fold-switching protein, and de novo-designed protein binders.

Our results show that AF3Score enhances success rates in de novo protein design compared to state-of-the-art methods. By assessing the input sequence against a provided structure rather than a predicted one, AF3Score provides a more robust estimation of the structural compatibility. This makes AF3Score a powerful tool for downstream screening in binder design, efficiently prioritizing candidates by assessing both monomer stability and target binding.

METHODS

Data Set of Protein–Protein Interactions. We selected 188 protein complexes from Docking Benchmark 5.5 that do not contain cofactors or ligands,¹⁸ of which 63 structures are antigen–antibody complexes. We used HDock to perform protein–protein docking simulations on the bound structures of these complexes. The docking process utilized HDock's default parameter settings, generating 100 candidate conformations for each complex. The quality of each docked pose was evaluated using DockQ v2,³⁶ focusing specifically on the interface DockQ scores between receptor and ligand chains.

From the 38 targets in CASP16's QMODEL,¹⁹ we selected 30 targets for testing, including 13 homo-oligomers and 17 hetero-oligomers. The selection criteria were: (1) targets with fewer than 3072 amino acids to avoid memory limitations, (2) targets with determined stoichiometry, and (3) targets with available EMA analysis results. We evaluated all models for the 30 targets, although only models with index 1 were considered for the QMODEL ranking.

Data Set of De Novo-Designed Binders. From a published data set of approximately 700,000 minibinders,³⁷ we extracted all 1,303 experimentally validated successful binders, and then randomly sampled 61,781 nonbinding designs to constitute our data set. For the ten minibinder structures with available crystal structures, we used HDock to generate 100 decoy structures for each and evaluated their interface quality using DockQ v2 scores between the designed binder and target protein chains.

Data Set of Fold Switching Proteins. We curated 45 fold-switching proteins from a dataset of 98 fold-switching proteins.²⁷ Among the original 98 proteins, 93 had their alternate conformations solved in the PDB and were considered in this study. For each fold-switching protein, we examined its dominant and alternative conformations, excluding those with insertions, deletions, or mutations in the fold-switching region to ensure identical sequences in this region between the two conformations, resulting in 45 proteins. The decoy generation was performed separately using both the dominant and the alternative conformations as references. For each target conformation, we generated 200 decoy structures using 3DRobot with an RMSD cutoff of 12Å.

3DRobot first identifies structure scaffolds from a non-redundant PDB library, then performs structure assembly simulations based on each scaffold, and finally selects decoys through a two-step energy minimization procedure. The MD ensembles of AdK were from a previous work.³³

Data Set of Protein–Ligand Interactions. We selected 178 protein–ligand complexes from the posebusters data set for our study.³⁸ To ensure data quality, we implemented a comprehensive preprocessing pipeline. First, we removed common crystallants (including SO₄, GOL, EDO, PO₄, ACT, PEG, etc.) from the original PDB files as these molecules are crystallization artifacts rather than natural binding partners. Subsequently, we standardized the structures by first converting ligands from SDF to PDB format, then standardizing ligand atom nomenclature according to the Chemical Component Dictionary and reassigning residue numbers and chain IDs. To generate decoy poses, we employed AutoDock Vina with an exhaustiveness of 32 and a search box of 15Å centered on the crystallographic ligand position, generating 50 docked poses for each complex. The LRMSD value of each docked pose was calculated relative to the pose of the native binding pose by DockQ v2.³⁶

ASSOCIATED CONTENT

Data Availability Statement

The code is available on GitHub <https://github.com/Mingchenchen/AF3Score/> and is installable. Full documentation complete with worked examples of how the package can be used are also available at github.

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jcim.5c00653>.

Spearman correlation and top-ranked metrics of AF3Score metrics for the Rosetta dataset; Pearson correlations between AF3Score metrics and DockQ scores on the Docking Benchmark 5.5 dataset, comprising 188 protein complexes; Spearman correlations between AF3Score metrics and TMScore or DockQ for multimer targets in QA; mean TMScore and DockQ for top-ranked models for each target selected by different metrics of AF3Score; performance comparison of monomer and complex-based metrics from AF3Score; computational efficiency of AF3Score; demonstration of the correlation between scores generated by AF3Score and their corresponding TMScores for target IWDV; Spearman correlations for antigen–antibody complexes ($N = 63$) and other types of PPIs ($N = 125$) in Docking Benchmark 5.5; correlation analysis; and other figures (PDF)

AUTHOR INFORMATION

Corresponding Author

Mingchen Chen – Changping Laboratory, Beijing 102206, China; orcid.org/0000-0003-3916-7871;
Email: mingchenchen@cpl.ac.cn

Authors

Yu Liu – Changping Laboratory, Beijing 102206, China
Qilin Yu – Changping Laboratory, Beijing 102206, China
Di Wang – Changping Laboratory, Beijing 102206, China

Complete contact information is available at:

<https://pubs.acs.org/doi/10.1021/acs.jcim.5c00653>

Author Contributions

M.C. designed research; Y.L. and D.W. performed research; Y.L., Q.Y., D.W., and M.C. analyzed data; and Y.L., Q.Y. and M.C. wrote the paper.

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

This work was supported by the start-up funding by Changping Laboratory.

REFERENCES

- (1) Watson, J. L.; et al. De Novo Design of Protein Structure and Function with RFDiffusion. *Nature* **2023**, *620*, 1089–1100.
- (2) Pacesa, M. et al. BindCraft: One-Shot Design of Functional Protein Binders. *bioRxiv* 2024.
- (3) Jumper, J.; et al. Highly Accurate Protein Structure Prediction with AlphaFold. *Nature* **2021**, *596*, 583–589.
- (4) Baek, M.; et al. Accurate Prediction of Protein Structures and Interactions Using a Three-Track Neural Network. *Science* **2021**, *373*, 871–876.
- (5) Abramson, J.; et al. Accurate Structure Prediction of Biomolecular Interactions with AlphaFold 3. *Nature* **2024**, *630*, 493–500.
- (6) Schafer, N. P.; Kim, B. L.; Zheng, W.; Wolynes, P. G. Learning To Fold Proteins Using Energy Landscape Theory. *Isr. J. Chem.* **2014**, *54*, 1311–1337.
- (7) Wolynes, P. G. Evolution, Energy Landscapes and the Paradoxes of Protein Folding. *Biochimie* **2015**, *119*, 218–230.
- (8) Chen, M.; Chen, X.; Schafer, N. P.; Clementi, C.; Komives, E. A.; Ferreira, D. U.; Wolynes, P. G. Surveying Biomolecular Frustration at Atomic Resolution. *Nat. Commun.* **2020**, *11*, 5944.
- (9) Roney, J. P.; Ovchinnikov, S. State-of-the-Art Estimation of Protein Model Accuracy Using AlphaFold. *Phys. Rev. Lett.* **2022**, *129*, No. 238101.
- (10) Bennett, N. R.; Coventry, B.; Goresnik, I.; Huang, B.; Allen, A.; Vafeados, D.; Peng, Y. P.; Dauparas, J.; Baek, M.; Stewart, L.; DiMaio, F.; De Munck, S.; Savvides, S. N.; Baker, D. Improving de Novo Protein Binder Design with Deep Learning. *Nat. Commun.* **2023**, *14*, 2625.
- (11) Park, H.; Bradley, P.; Greisen, P.; Liu, Y.; Mulligan, V. K.; Kim, D. E.; Baker, D.; DiMaio, F. Simultaneous Optimization of Biomolecular Energy Functions on Features from Small Molecules and Macromolecules. *J. Chem. Theory Comput.* **2016**, *12*, 6201–6212.
- (12) Alford, R. F.; et al. The Rosetta All-Atom Energy Function for Macromolecular Modeling and Design. *J. Chem. Theory Comput.* **2017**, *13*, 3031–3048.
- (13) Hiranuma, N.; Park, H.; Baek, M.; Anishchenko, I.; Dauparas, J.; Baker, D. Improved Protein Structure Refinement Guided by Deep Learning Based Accuracy Estimation. *Nat. Commun.* **2021**, *12*, 1340.
- (14) Yan, Y.; Tao, H.; He, J.; Huang, S.-Y. The HDock Server for Integrated Protein–Protein Docking. *Nat. Protoc.* **2020**, *15*, 1829–1852.
- (15) Honorato, R. V.; Trellet, M. E.; Jiménez-García, B.; Scharschmidt, J. J.; Giuliani, M.; Reys, V.; Koukos, P. I.; Rodrigues, J. P. G. L. M.; Karaca, E.; Van Zundert, G. C. P.; Roel-Touris, J.; Van Noort, C. W.; Jandová, Z.; Melquiond, A. S. J.; Bonvin, A. M. J. J. The HADDOCK2.4 Web Server for Integrative Modeling of Biomolecular Complexes. *Nat. Protoc.* **2024**, *19*, 3219–3241.
- (16) Vajda, S.; Hall, D. R.; Kozakov, D. Sampling and Scoring: A Marriage Made in Heaven: Sampling and Scoring. *Proteins: Struct., Funct., Bioinf.* **2013**, *81*, 1874–1884.
- (17) Cao, Y.; Shen, Y. Energy-based Graph Convolutional Networks for Scoring Protein Docking Models. *Proteins: Struct., Funct., Bioinf.* **2020**, *88*, 1091–1099.
- (18) Vreven, T.; Moal, I. H.; Vangone, A.; Pierce, B. G.; Kastiris, P. L.; Torchala, M.; Chaleil, R.; Jiménez-García, B.; Bates, P. A.; Fernandez-Recio, J.; Bonvin, A. M.; Weng, Z. Updates to the Integrated Protein–Protein Interaction Benchmarks: Docking Benchmark Version 5 and Affinity Benchmark Version 2. *J. Mol. Biol.* **2015**, *427*, 3031–3041.
- (19) Home - CASP16. <https://predictioncenter.org/casp16/index.cgi>.
- (20) Lim, D.; Park, H. U.; De Castro, L.; Kang, S. G.; Lee, H. S.; Jensen, S.; Lee, K. J.; Strynadka, N. C. J. Crystal Structure and Kinetic Analysis of β -Lactamase Inhibitor Protein-II in Complex with TEM-1 β -Lactamase. *Nat. Struct. Biol.* **2001**, *8*, 848–852.
- (21) Blech, M.; et al. One Target—Two Different Binding Modes: Structural Insights into Gevokizumab and Canakinumab Interactions to Interleukin- β . *J. Mol. Biol.* **2013**, *425*, 94–111.
- (22) Baek, M.; McHugh, R.; Anishchenko, I.; Jiang, H.; Baker, D.; DiMaio, F. Accurate Prediction of Protein–Nucleic Acid Complexes Using RoseTTAFoldNA. *Nat. Methods* **2024**, *21*, 117–121.
- (23) Watson, J. L. et al. Broadly Applicable and Accurate Protein Design by Integrating Structure Prediction Networks and Diffusion Generative Models. *bioRxiv* 2022.
- (24) Adaptiv Foundry. <https://foundry.adaptivbio.com/competition>.
- (25) Porter, L. L.; Looger, L. L. Extant Fold-Switching Proteins Are Widespread. *Proc. Natl. Acad. Sci. U. S. A.* **2018**, *115*, 5968–5973.
- (26) Schafer, J. W.; Porter, L. L. Evolutionary Selection of Proteins with Two Folds. *Nat. Commun.* **2023**, *14*, 5478.
- (27) Porter, L. N. E. AlphaFold2 Fails to Predict Protein Fold Switching. *Protein Sci.* **2022**, *31*, No. e4353.
- (28) Chakravarty, D.; Schafer, J. W.; Chen, E. A.; Thole, J. F.; Ronish, L. A.; Lee, M.; Porter, L. L. AlphaFold Predictions of Fold-Switched Conformations Are Driven by Structure Memorization. *Nat. Commun.* **2024**, *15*, 7296.
- (29) Evans, R. et al. Protein Complex Prediction with AlphaFold-Multimer. *bioRxiv* 2021.
- (30) Stein, R. A.; Mchaourab, H. S. SPEACH_AF: Sampling Protein Ensembles and Conformational Heterogeneity with AlphaFold2. *PLOS Computational Biology* **2022**, *18*, No. e1010483.
- (31) Wayment-Steele, H. K.; Ojoawo, A.; Otten, R.; Apitz, J. M.; Pitsawong, W.; Hömberger, M.; Ovchinnikov, S.; Colwell, L.; Kern, D. Predicting Multiple Conformations via Sequence Clustering and AlphaFold2. *Nature* **2024**, *625*, 832–839.
- (32) Kalakoti, Y.; Wallner, B. AFsample2: Predicting Multiple Conformations and Ensembles with AlphaFold2. *Commun. Biol.* **2024**, *8*, 373.
- (33) Guan, X.; Tang, Q.-Y.; Ren, W.; Chen, M.; Wang, W.; Wolynes, P. G.; Li, W. Predicting Protein Conformational Motions Using Energetic Frustration Analysis and AlphaFold2. *Proc. Natl. Acad. Sci. U. S. A.* **2024**, *121*, No. e2410662121.
- (34) Chang, Y.-G.; et al. A Protein Fold Switch Joins the Circadian Oscillator to Clock Output in Cyanobacteria. *Science* **2015**, *349*, 324–328.
- (35) Schafer, J. W.; Porter, L. L. AlphaFold2's Training Set Powers Its Predictions of Fold-Switched Conformations. *bioRxiv* 2024.
- (36) Mirabello, C.; Wallner, B. DockQ v2: Improved Automatic Quality Measure for Protein Multimers, Nucleic Acids, and Small Molecules. *Bioinformatics* **2024**, *40*, No. btad586.
- (37) Cao, L.; et al. Design of Protein-Binding Proteins from the Target Structure Alone. *Nature* **2022**, *605*, 551–560.
- (38) Buttenschon, M.; Morris, G. M.; Deane, C. M. PoseBusters: AI-based Docking Methods Fail to Generate Physically Valid Poses or Generalise to Novel Sequences. *Chemical Science* **2024**, *15*, 3130–3139.